# DIFFERENTIAL EQUATIONS - MATH 251

## FIRST ORDER ODE

**M. A. BOATENG (Ph.D., MIMA)**



*KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY*

# TABLE OF CONTEXT

o


- ▶ EXACTNESS
- ▶ NON-EXACT ODES
- ▶ EQUATIONS REDUCIBLE TO THE LINEAR FORM (BERNOULLI)
- ▶ RICCATI'S EQUATION



# Table of Contents

1 EXACTNESS


► EXACTNESS

► NON-EXACT ODES

► EQUATIONS REDUCIBLE TO THE LINEAR FORM (BERNOULLI)

► RICCATI'S EQUATION



# EXACTNESS

## 1 EXACTNESS

Given f(x,y)=c

$\frac{df}{dt} = \frac{\delta f}{\delta x}\frac{dx}{dt} + \frac{\delta f}{\delta y}\frac{dy}{dt}$

$df = \frac{\delta f}{\delta x}dx + \frac{\delta f}{\delta y}dy = 0 - - - (1)$

$M(x,y)dx + N(x,y)dy = 0 - - - (2)$

$M(x,y) = \frac{\delta f}{\delta x}, N(x,y) = \frac{\delta f}{\delta y}$

If $\frac{\delta M(x,y)}{dy} = \frac{\delta N(x,y)}{dx}$, $\exists$ a function f(x,y)=c that solves $M(x,y)dx + N(x,y)dy = 0$, then this is an **Exact D.E**



# EXAMPLE

## 1 EXACTNESS

1. Solve $(1-2xy)dx+(4y^3-x^2)dy=0$

**SOLUTION**

Let f(x,y)=c be the solution

$M(x,y)=\frac{\delta f}{\delta x}=1-2xy$

$N(x,y)=\frac{\delta f}{\delta y}=4y^3-x^2$ $\frac{\delta M(x,y)}{dy}=-2x$

$\frac{\delta N(x,y)}{dx}=-2x$, hence exact

$f(x,y)=\int M(x,y)dx+h(y)$

$$f(x,y) = \int (1 - 2xy)dx + h(y)$$
$$= x - x^2y + h(y)$$

But $\frac{\delta f(x.y)}{\delta y} = \frac{\delta(x - x^2y + h(y))}{\delta y} = N(x,y)$

$$\implies -x^2 + g^l(y) = 4y^3 - x^2$$
$$h^l(y) = 4y^3$$
$$h(y) = y^4 + c$$
$$f(x,y) = x - x^2y + y^4 + C$$

## TRIALS

Solve the following ODEs using the method of exactness

1. $(ycosx + 2xe^y)dx + (sinx + x^2e^y + 2)dy = 0$
2. $(3xy + y^2) + (x^2 + xy)y' = 0$
3. $2xy^2 + 4 = 2(3 - x^2y)\frac{dy}{dx}$



# Table of Contents

2 NON-EXACT ODES


- EXACTNESS
- NON-EXACT ODES
- EQUATIONS REDUCIBLE TO THE LINEAR FORM (BERNOULLI)
- RICCATI'S EQUATION



# NON-EXACT ODES

## 2 NON-EXACT ODES

Given that $M(x,y)dx + N(x,y)dy = 0$ is not Exact. Suppose that, $\exists$ a $\mu = \mu(x.y)$ s.t

$\mu M(x,y)dx + \mu N(x,y)dy = 0$ is Exact. [$\mu(x,m)$ = integrating factor].

$$\implies \frac{\delta(\mu M)}{dy} = \frac{\delta(\mu N)}{dx}$$

$$\mu(x) = e^{\int(\frac{M_y - N_x}{N})dx}$$

Here $\mu(x)$ is chosen on the assumption that the integrating factor is a function of **x** only.

Example:
Solve the ODE given $(3xy + y^2)dx + (x^2 + xy)dy = 0$



# SOLUTION

## 2 NON-EXACT ODES

The equation is not exact hence let $\mu(x) = e^{\int(\frac{M_y - N_x}{N})dx}$

$M_y = 3x + 2y, N_x = 2x + y$

$\mu(x) = e^{\int(\frac{3x+2y-2x-y}{x^2+xy})dx}$

$\implies \mu(x) = e^{\int \frac{1}{x}dx} = e^{\ln x} = x$

We have $(3x^2y + xy^2)dx + (x^3 + x^2y)dy = 0$

$\frac{\delta M}{\delta y} = 3x^2 + 2xy$ and $\frac{\delta N}{\delta x} = 3x^2 + 2xy$

$f(x, y) = \int M(x, y)dx$ OR $\int N(x, y)dy$

$f(x, y) = x^3y + \frac{x^2y^2}{2} + c$ OR $x^3y + \frac{x^2y^2}{2} + c$

$x^3y + \frac{x^2y^2}{2} = c$

## TRIAL

Solve the following non-exact differential equations.

1. $(3xy - y^2)dx + x(x - y)dy = 0$
2. $y(1 - xy)dx + xdy = 0$
3. $(3y^2 + 4x)dx + 3xydy = 0$



# Table of Contents

3 EQUATIONS REDUCIBLE TO THE LINEAR FORM (BERNOULLI)


► EXACTNESS

► NON-EXACT ODES

► EQUATIONS REDUCIBLE TO THE LINEAR FORM (BERNOULLI)

► RICCATI'S EQUATION



# EQUATIONS REDUCIBLE TO THE LINEAR FORM (BERNOULLI)



Given $y^{'} + p(x)y = q(x)y^{n}$

$y'y^{-n} + p(x)y^{1-n} = q(x)$

Let $z = y^{1-n}$

$z^{'} = (1-n)y^{-n}y'$

$\frac{z^{'}}{1-n} + p(x)z = q(x)$

$z^{'} + (1-n)p(x)z = (1-n)q(x)$

**Example**

$x^2y - x^3\frac{dy}{dx} = y^4 cosx$



**SOLUTION**

$$-x^2y + x^3\frac{dy}{dx} = -y^4 cosx$$

$$\frac{dy}{dx} - \frac{1}{x}y = -\frac{cosx}{x^3}y^4$$

from $z' + (1-n)p(x)z = (1-n)q(x)$

$$z' + \frac{3}{x}z = \frac{3cosx}{x^3}$$

$$\mu(x) = e^{3\int \frac{1}{x}dx}$$

$$\mu(x) = x^3$$

$$x^3z' + 3x^2z = -3cosx$$

$x^3z = 3sinx + c$

but $z = y^{-3}$

$x^3y^{-3} = 3sinx + c$

$y^3 = \frac{x^3}{3sinx+c}$

## TRIAL

Solve the following Bernoulli differential equations:

1. $t^2y' + 2ty - y^3 = 0$
2. $y' + xy = xy^2$
3. $xy' + y = \frac{1}{y^2}$



# Table of Contents

4 RICCATI'S EQUATION

- EXACTNESS
- NON-EXACT ODES
- EQUATIONS REDUCIBLE TO THE LINEAR FORM (BERNOULLI)
- RICCATI'S EQUATION

$\frac{dy}{dx} = P(x) + Q(x)y + R(x)y^2$

Given $y_1 = t$ as a particular solution to the differential equation,

let $z = \frac{1}{y-t}$

$y = t + \frac{1}{z}$

**EXAMPLE**

Solve $\frac{dy}{dx} = -2 - y + y^2$

$y_1 = 2$ is a particular solution, find the general solution.



**SOLUTION**

let $z = \frac{1}{y-2}$

$\implies y = 2 + \frac{1}{z}$

$y' = -\frac{z'}{z^2}$

$-\frac{z'}{z^2} = -2 - (2 + \frac{1}{z}) + (2 + \frac{1}{z})^2$

$-z' = -2z^2 - z(2z + 1) + (2z + 1)^2$

$z' = -3z - 1$ (Now linear)

$$z' + 3z = -1$$

$$\mu(x) = e^{\int 3dx}$$

$$\mu(x) = e^{3x}$$

$$\int d(e^{3x}z) = -\int e^{3x}dx$$

$$ze^{3x} = -\frac{1}{3}e^{3x} + C$$

$$z = -\frac{1}{3} + Ce^{-3x}$$

$$y = \frac{-6Ce^{3x}-1}{1-3Ce^{3x}}$$

## EXERCISE

Solve the following Riccati equations;

1. $\frac{dy}{dx} = \frac{y^2}{x} + \frac{y}{x} + x$, given $y_1 = 2$ is a particular solution, find the general solution.
2. $y' = -2 + y + y^2$



# DIFFERENTIAL EQUATIONS - MATH 251

*Thank you for listening!*
*Any questions?*


thank you!